

DL3004 Final Project Overview

Complete Data Pipeline

Project Introduction

The DL3004 Final Project is an end-to-end Data Warehousing and Business Intelligence implementation. Students will not only create tables and dashboards but will build a complete analytical data pipeline similar to a real enterprise environment.

The complete pipeline expected in this project:

Source Systems → Data Ingestion → Staging Layer → Data Cleaning & Transformation → Data Warehouse → OLAP Analysis → BI Reporting → Business Decisions.

Project Scenario Example

Example: CampusCart is an online retail company. Management wants to understand sales performance, customer behavior, inventory movement, and promotion effectiveness.

Students will build a warehouse that answers questions such as:

- Which products generate the highest revenue?
- Which customer segments are most valuable?
- Which regions are declining?
- Which promotions increase sales?

Technology Stack

Free Implementation:

- PostgreSQL
- pgAdmin
- Python/Pandas
- Power BI Desktop
- Draw.io
- GitHub

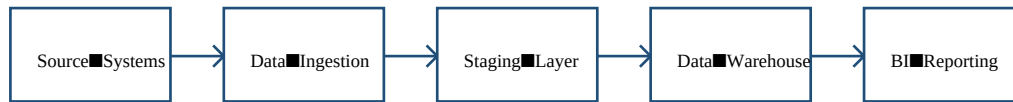
Optional Industry Tools:

- Snowflake
- BigQuery Sandbox
- Databricks Community Edition
- dbt Core

Students choosing cloud tools must still maintain reproducible SQL/scripts.

Expected Final Architecture

Example architecture showing movement of data from raw sources to business insight.



Timeline and Marks

Phase 1 (Weeks 1-3): Design - 30%

Phase 2 (Weeks 4-6): Implementation - 40%

Phase 3 (Weeks 7-8): Analytics and BI - 30%

Total Project Weight: 15%

Suggested Final Repository Structure

DWBI-Final-Project

01_Project_Documentation

■■■ Requirements.pdf

■■■ Schema.pdf

■■■ Data Dictionary.xlsx

02_Data

■■■ raw_data

■■■ cleaned_data

03_ETL

■■■ staging.sql

■■■ dimensions.sql

■■■ facts.sql

04_Data_Warehouse

■■■ warehouse.sql

05_Analytics

■■■ OLAP.sql

■■■ Views.sql

06_Dashboard

■■■ PowerBI.pbix

README.md

Project Outcome

This gives students a complete portfolio-quality DWBI project while requiring zero paid infrastructure. It maps naturally to the course journey: DW foundations → dimensional modeling → ETL → OLAP → BI reporting.